



Evaluating head impact intensities and accelerations using flexible wearable sensors for traumatic brain injury assessment

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ABSTRACT

In response to the significant health threat posed by Traumatic Brain Injury (TBI) resulting from sudden collisions on the head, there is an urgent need for timely assessment and evaluation. With the wide applications of wearable devices in health monitoring, here we report a flexible and wearable smart headband based on a polyvinylidene fluoride (PVDF) sensor to evaluate the impact force and acceleration for TBI assessment. Based on the electromechanical coupling characteristics of PVDF, a lumped parameters model of an equivalent circuit was obtained by a system identification method, which accurately transforms the output signal of the PVDF sensor to the impact force for TBI assessment of focal brain damage. Besides, a drop tower was used to impose different impact intensity levels on a dummy head wearing the smart headband; the results (MSE < 0.018) of transformed impact force show the universal applicability of the model. Moreover, the correlation analysis results show that the impact force has a positive correlation ($R > 85\%$) with impact acceleration at the head center of gravity, which enables the TBI assessment of diffuse brain damage. This smart headband can be used for real-time monitoring of human head impact and has excellent potential for application in medical and sports devices.

1. Introduction

As one of the most severe diseases with the highest mortality and disability rates worldwide, traumatic brain injury (TBI) caused by various sudden collisions threatens people's lives and health [1]. In addition to the immediate trauma to the brain, TBI may also have irreversible effects on the patient's linguistic competence, mobility, and intelligence [2]. More than 50 million people have TBIs each year worldwide, and it is estimated that about half the world's population will have one or more TBIs over their lifetime [3,4]. The inability to obtain timely and accurate on-site medical TBI data is one of the essential reasons that patients miss the golden treatment period [5]. Accordingly, real-time detection of TBI is paramount, as it enables timely assessment of brain injuries, thereby mitigating the progression to more severe harm.

Given the critical necessity for real-time monitoring and evaluation of head impacts, wearable devices have emerged as innovative solutions in the assessment of TBI [6,7]. Compared to hospital-based evaluations

of patients, in-situ physiological data generated during the impact process can be collected by wearable devices, which provide a more comprehensive basis for evaluating brain injury. The causes of Brain damage in TBI are classified as (a) focal brain damage due to impact contact injury types resulting in contusion, laceration, and intracranial hemorrhage or (b) diffuse brain damage due to acceleration/deceleration injury types resulting in diffuse axonal injury or brain swelling [8, 9]. In subsequent studies, inertial sensors and pressure sensors have been utilized in wearable medical devices for the assessment of head impacts. Liu et al. [10] have developed an instrumented mouthguard with built-in inertial sensors and demonstrated its applicability in detecting head collisions in football sports. Nicholas et al. [11] have incorporated the SIM-G head impact sensor into headbands, evaluating the accuracy of this sensor in measuring linear acceleration and angular velocity, thereby substantiating its application in monitoring head impacts in water polo and terrestrial sports. However, wearable head collision detection devices based on inertial sensors primarily focus on measuring parameters of inertial forces exerted on the head, lacking the

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ability to provide direct information about the impact force of collisions.

Given that head impacts last between 5.5 and 13.7 ms [12], wearable devices for impact detection put forward requirements for the rapid response speed for sensors. Hence, flexible piezoelectric sensors, owing to their advantages, such as good conformity with the human body and excellent dynamic response capabilities, are highly suitable for applications in the field of impact detection [13,14]. Within the piezopolymers, polyvinylidene fluoride (PVDF) films have been widely investigated as sensor materials due to their chemical stability, high piezoelectric, and high flexibility [15]. Previous research has shown that the output voltage from a PVDF piezoelectric sensor deployed in helmets can be converted into the magnitude of head impact pressure using a simplified first-order equivalent model [16]. However, this method neglects the dynamic response characteristics of the sensor during the impact and fails to incorporate a thorough analysis of the impact acceleration, which fails to provide an assessment of TBI requiring comprehensive information for impact.

This paper proposes a flexible, wearable smart headband based on PVDF piezoelectric film to evaluate the impact force and acceleration during head impacts. To determine the applicability of the PVDF sensor in evaluating different impact levels, a drop tower equipped with a dummy head was designed to simulate different intensities of head impact, which enables the acquisition of PVDF signal voltage, impact force, and impact acceleration through experiments. Subsequently, the lumped parameter model, which combines the electrical and mechanical characteristics of the PVDF sensor, was extracted using a system identification method to ensure the precise evaluation of the impact force for TBI assessment of focal brain damage. Based on the transformation results, a correlation between acceleration and impact force was established, which enables the estimation of impact acceleration for TBI assessment of diffuse brain damage. Therefore, the smart headband exhibits effective performance in monitoring the head impact, proposing a new solution for the real-time assessment of TBI.

2. Methods

2.1. Fabrication of the PVDF-based smart headband

The fabrication process of the flexible PVDF sensor is illustrated in Fig. 1a. At first, a PVDF film (thickness of 0.1 mm) with Aluminum films as electrodes covering both sides of the PVDF was fabricated by KEXIN ELECTRONIC, taken with a dimension of 30 mm × 30 mm. Then, the wire was attached to Aluminum films using two conductive copper tapes (5 mm × 5 mm) to ensure good contact between the wire and the electrode. After that, the PVDF film containing the electrodes and wires was encapsulated with two polyimide (PI) films to provide the PVDF film with mechanical robustness [17]. Combining the flexible PVDF sensor with a headband made of fabric, a PVDF-based smart headband (as shown in Fig. 1b) was proposed for evaluating the head impact, which can adapt to different head circumferences due to its elasticity. Moreover, the modular design of this smart headband allows for integration into various types of helmets, ensuring its function in assessing Traumatic Brain Injury across contact processes such as hockey, football, etc, thereby demonstrating its wide adaptability. When mechanical impacts are applied to the smart headband, it results in corresponding signal voltages being output by the PVDF sensor, which enables the evaluation of the impact force and the estimation acceleration respectively.

2.2. Lumped parameters modeling of the PVDF sensor

The critical element in the accurate assessment of head impacts lies in the precise representation of head impact force through the signal voltage generated by the flexible PVDF sensor. In the electrical domain, PVDF film behaves like a capacitive generator-type device, giving an electric output signal without any external electrical excitation [18]. Eq.1 indicates the correspondence between signal voltage V and impact force F :

$$V = \frac{1}{C_s} Q = \frac{d_{33}}{C_s} F \quad (1)$$

Where d_{33} is the piezoelectric coefficient for PVDF, C_s is the capacitance

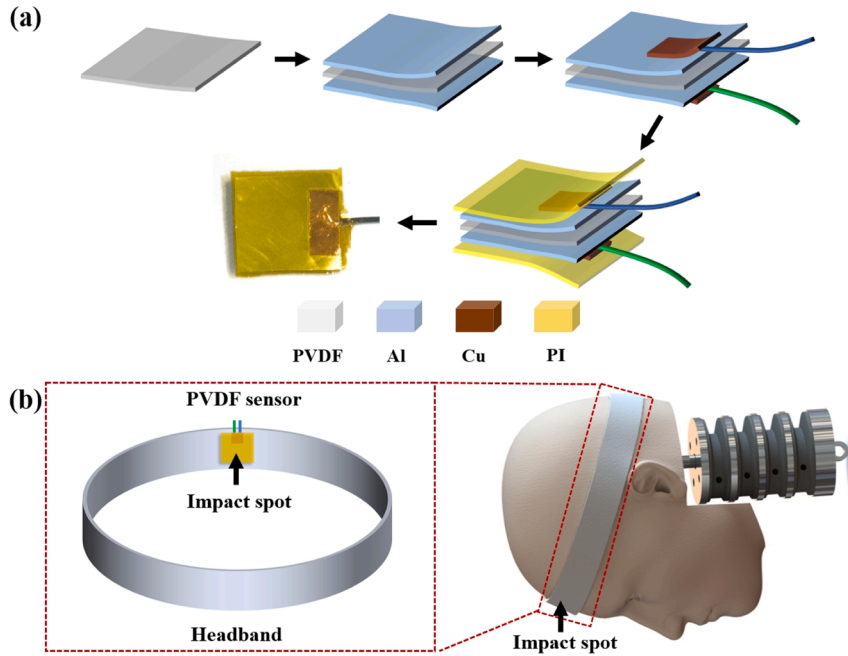


Fig. 1. Fabrication of the PVDF-based smart headband. (a) Schematic illustration of the fabrication process of the flexible PVDF sensor (area of 30 mm × 30 mm). (b) The structure of the PVDF-based smart headband (including flexible PVDF sensor and headband) and its placement on the head demonstrated the manner and position of the sensor applied to the headband.

of the PVDF film. According to Eq. 1, the impact force F can be determined by combining the piezoelectric coefficient d_{33} , the capacitance C_s , and the signal voltage V . As mentioned in Eq. 1, the characteristics of piezopolymer films can be described by capacitors for low-frequency impacts. However, due to the viscoelastic characteristics of polymer films under high-frequency impacts, more complex models are required for higher frequencies [19]. In addition, the evaluation of TBI using the Head Injury Criterion (HIC), including the integral of impact signal [20], necessitates the determination of the complete time-domain signal in the impact event. Therefore, since the inability of Eq. 1 to reproduce the complete time domain of impact force needed in TBI assessment of focal brain damage, it is essential to utilize the electromechanical properties and piezoelectric effect of the PVDF film to enhance the detection capability of the flexible wearable device for impact forces [21]. Eq. 2 shows the lumped parameter model for PVDF film [22], which was developed to describe the interaction between signal voltage V and impact force F :

$$V = (k_\varphi, C_1, L_1, R_1, C_s, R_s) F \quad (2)$$

In the mechanical domain, the model of PVDF film is described as a simple spring-mass-damper [22]. The spring is presented by a capacitor C_1 ; the mass is presented by an inductor L_1 , which represents the mechanical inertia of the film; and the damper is defined by the resistance R_1 , which represents the mechanical damping forces that arise from the viscosity of the film material and air. The spring-mass-damper model is transformed into an equivalent circuit in the mechanical domain and characterized by an RLC model, as shown in Fig. 2a. In the electrical domain, the model of PVDF film is considered here as a parallel-plate capacitor with a capacitance C_s and a resistance R_s as shown in Fig. 2a. The capacitance C_s is defined by the thickness of the PVDF film, electrode area, and the permittivity of the PVDF; the resistance R_s is determined by the parallel connection within the internal resistance of the film and the external resistance of the measurement circuit. As shown in Fig. 2a, the transformer indicates conversion between the mechanical and electrical domains, defined by the electromechanical coupling constant k_φ , which measures the conversion efficiency between electrical and mechanical energy [23]. Finally, the equivalent circuit (shown in Fig. 2a) is simplified to the electrical circuit (shown in Fig. 2b), aiming to obtain the relationship between signal voltage V and impact force F . A transfer function is established based on the simplified equivalent electrical circuit with the parameters (C_1, L_1, R_1, C_s, R_s) mentioned above:

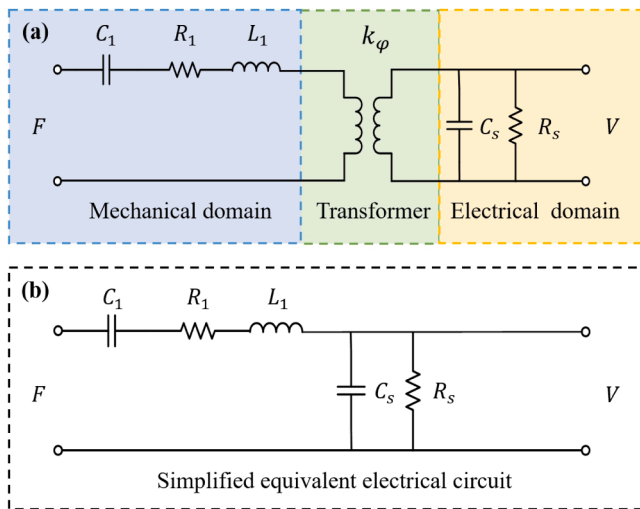


Fig. 2. (a) The equivalent circuit with lumped parameters in the mechanical domain (C_1, L_1, R_1), the transformer (k_φ), and the electrical domain (C_s, R_s). (b) The simplified equivalent electrical circuit.

$$G(s) = \frac{V(s)}{F(s)} = \frac{s}{as^3 + bs^2 + cs + d} \quad (3)$$

Where $G(s)$ is the transfer function, $V(s)$ is the signal voltage in the complex field, $F(s)$ is the impact force, respectively. Besides, the parameters in the transfer function are defined as follows: $a = L_1 C_s$, $b = L_1/R_s + R_1 C_s$, $c = R_1/R_s + C_s/C_1 + 1$, $d = 1/(C_1 R_s)$.

2.3. Experimental setup

The impact tests were carried out on an experimental setup, which is presented in Fig. 3a, consisting of three components: (1) Head drop tower, (2) Dummy head, and (3) Data acquisition module. Testing was conducted using a dummy head dropping from a pre-set height onto the ground to obtain the impact result through the test. To simulate different head impact levels, the head drop tower is constructed using the telescopic tube measuring approximately 100 cm for connection between the track frame and the dummy head, which implemented impact by the specified drop height from 0 to 100 cm. Besides, the drop tower includes sliding modules for screw connection between the track frame and the dummy head. The Hybrid III 50th Male is selected as the dummy head with a weight of 6.08 kg, which is the standard test dummy for frontal crash tests (shown in Fig. 3a), and has a similar biomechanical response and impact characteristics to humans when it contacts with the impact sensor (shown in Fig. 3a). In addition, an accelerometer is installed at specific positions inside the dummy's head (shown in Fig. 3c), which can be used to measure the acceleration in the center of gravity in the dummy head for TBI assessment of diffuse brain damage during the impact process.

The design of the data acquisition module includes (1) the wearable device acquisition module, (2) the impact force acquisition module, and (3) the acceleration acquisition module, as shown in Fig. 3a. The PVDF-based smart headband is fixed on the dummy head (shown in Fig. 3a), with the PVDF sensor located at the point of impact between the head and the ground. When an impact occurs, the signal voltage generated by the smart headband is acquired by a data acquisition instrumentation (NI-DAQ, National Instruments, Inc.) and then recorded on a laptop at a sampling rate of 5 kHz. A larger impact will cause the PVDF film to produce an instantaneous excessive response, and the voltage will exceed the input limit of NI-DAQ (± 10 V), which may lead to data loss. Therefore, the acquisition circuit includes a voltage follower (shown in Fig. 3e) based on op-amp (LM324, Texas Instruments, Inc.) for buffering isolation and a bias resistor of 20 k Ω , which is used for dividing the voltage generated by wearable devices. The impact force acquisition module is composed of a quartz piezoelectric sensor (6012, Lindmc, Inc.) installed at the bottom of the head drop tower (shown in Fig. 3b) for calibrating impact force and its accompanying charge amplifier (6901, Lindmc, Inc.). The signal output by the impact force acquisition module is 0–10 V, corresponding to a voltage response of 0–10 kN force. Similar to the smart headband acquisition method, NI-DAQ was used to collect the voltage of impact force signals and then recorded on a laptop at a sampling rate of 5 kHz. To measure the impact accelerations during the test, an accelerometer (ADXL377, Analog Devices, Inc.) with a full-scale range of ± 200 g was mounted onto the 3D-printed stand fixed at the center of gravity in the dummy head, which collects the acceleration at a sampling rate of 500 Hz through the Arduino IO port and transmits it to the PC through serial communication.

3. Results and discussion

3.1. Identification of lumped parameters

During the test, the dummy head was released from specified heights (10 cm, 20 cm, 30 cm, 40 cm, 50 cm), causing it to experience impacts of different levels with the ground. The data acquisition module collects

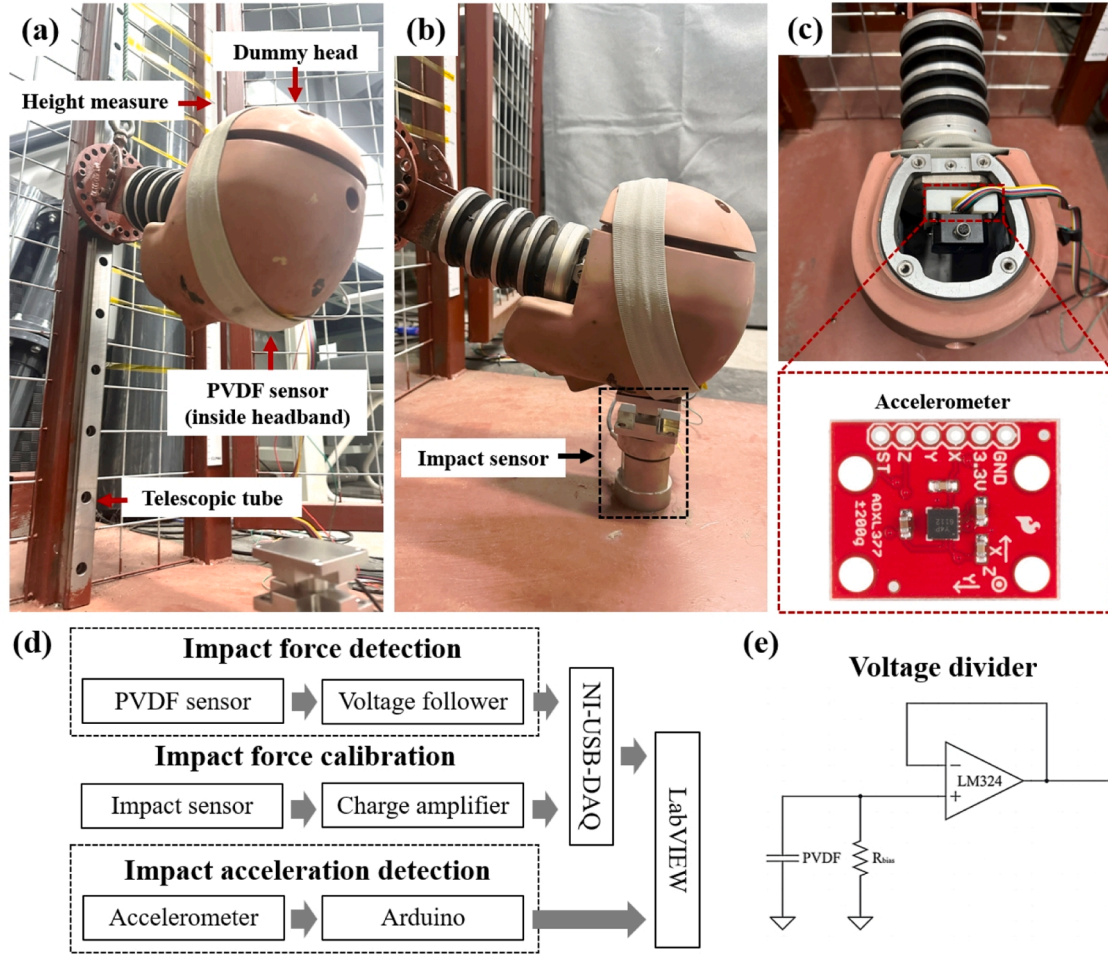


Fig. 3. (a) Photographs of the side view of the dummy head and drop tower (b) Photos of dummy head contact to the impact sensor (c) Installation method of accelerometer inside the dummy head. (d) Lateral view illustration of the measurement process with hardware connections. (e) Schematic of op-amp-based amplification circuit based on LM324 for buffering isolation.

the signal voltage of PVDF, impact force, and impact acceleration during the entire impact process. Due to environmental noise, a high-pass filter at the cutoff frequency of 50 Hz is used to filter power-line interference. In addition to facilitating data processing, all data is sliced into segments with 200 sampling points (5 kHz, 0.04 s) to include the collision process, considering that the collision process occurs within 0.02 s for different heights of impact data.

Firstly, to obtain the relationship between impact force F and signal voltage V , the system identification method [24] is used to determine the parameters in the transfer function $G(s)$ in Eq. 3. The parameters of both the numerator and denominator polynomials were estimated from the experimental data including signal voltage and impact force by utilizing a Gauss-Newton least squares search fitting algorithm. The optimization of parameters within the transfer function was based on a dataset comprising 50 sets of experiments (divided into 35 sets for identification and 15 sets for validation) under varying drop heights (10 cm, 20 cm, 30 cm, 40 cm, and 50 cm, with 10 trials for each height), thereby enabling an accurate transformation from the equivalent electrical model to different impact levels. For example, the results (in three different drop heights of 10 cm, 30 cm, and 50 cm) of the system identification method, with impact force F as the input and signal voltage V as the output, show the transformations of the model for different drop heights in Fig. 4a, b, c, respectively. In order to make the comparison of transfer function parameter fitting results more intuitive, the parameters are transformed into dimensionless form in transfer function parameter identification. The transfer function parameters identification results

illustrate the reliability of the model based on fitted parameters under various impact levels, which maintained small deviations and consistency at different fall heights, as shown in Fig. 4d. Therefore, the identification result of the transfer function parameters, i.e., $a = 2.76 \times 10^{-8}$, $b = 7.36 \times 10^{-4}$, $c = 1.59$, $d = 3.59 \times 10^3$, is obtained from identification sets (35 sets of experiments) was averaged and the identified transfer function based on the simplified equivalent circuit is obtained as:

$$G(s) = \frac{V(s)}{F(s)} = \frac{s}{2.76 \times 10^{-8}s^3 + 7.36 \times 10^{-4}s^2 + 1.59s + 3.59 \times 10^3} \quad (4)$$

A set with a drop height of 30 cm demonstrates the model transformation results of the equivalent electrical model with the comparison between the transformed voltage V_t and signal voltage V , as shown in Fig. 4e. Additionally, the Square Error (SE) of the model transformation results is shown in Fig. 4f, which demonstrates the high correlation between the time domain of transformed voltage V_t and signal voltage V . The identified transfer function model was then validated against validation sets (15 sets of experiments), and its performance metric, such as MSE, was evaluated. The verification result shows that the mean identification error of the transfer function model between the transformed voltage V_t and signal voltage V is $\text{MSE} = 0.000538$, illustrating the reliability of system identification of the equivalent electrical circuit model.

After determining the parameters of the transfer function, the transformation of the impact force F to the PVDF sensor output signal voltage V was theoretically accomplished based on the equivalent

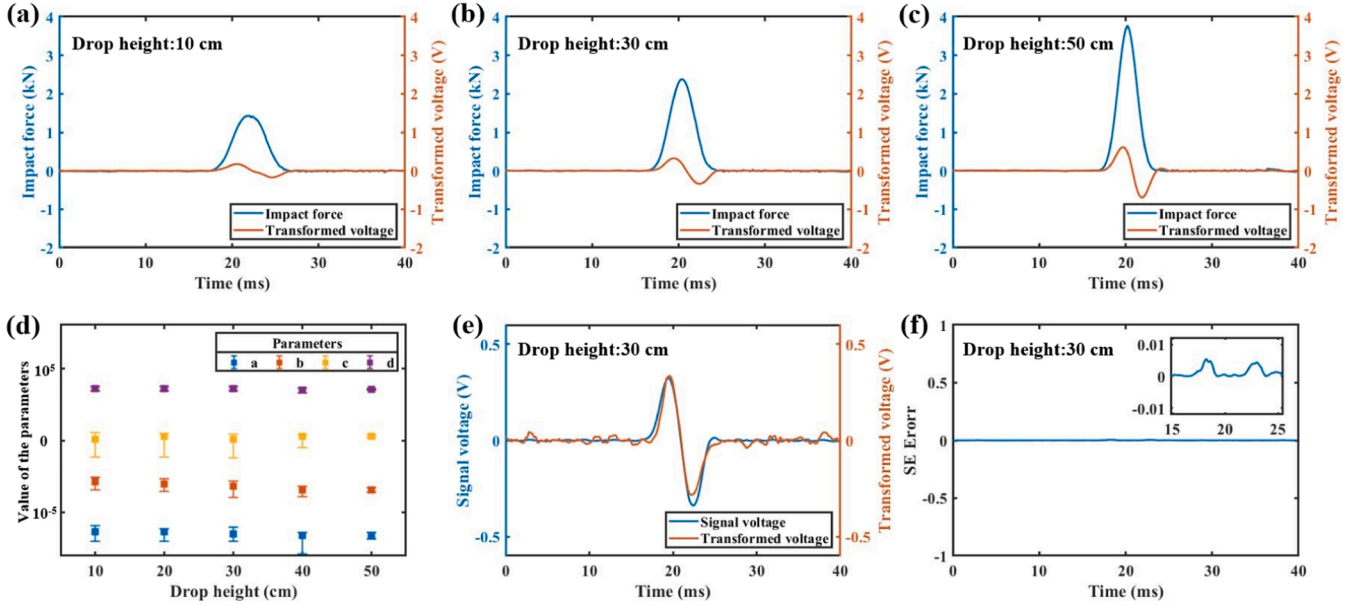


Fig. 4. Comparison of three examples between impact force F and transformed voltage V_t (a) drop height of 10 cm, (b) drop height of 30 cm, (c) drop height of 50 cm, respectively. (d) Values of parameters (in dimensionless form) extraction in the transfer function. ($a = L_1 C_s$, $b = L_1/R_s + R_1 C_s$, $c = R_1/R_s + C_s/C_1 + 1$, $d = 1/(C_1 R_s)$) (e) Comparison of an example (drop height of 30 cm) between signal voltage V and transformed voltage V_t . (f) SE error of an example (drop height of 30 cm) between signal voltage V and transformed voltage V_t .

electrical circuit model. Furthermore, to achieve a proper response of the sensor output signal voltage V to impact forces F , it is essential to inverse the identified system and its transfer function $G(s)$. However, it can be observed from the transfer function that directly inverting the system will swap the poles and zeros of its transfer function, leading to potential instability in the inverse system with transfer function $1/G(s)$. Based on the above problems, the non-linear least squares optimization algorithm was utilized to estimate the actual impact force F on the head by inversely solving the transformed force F_t as the input of the estimated transfer function based on the known signal voltage V as the output of the system. In essence, the process of solving the transformed force F_t in reverse is an optimization problem in which the waveform of the transformed force F_t is continuously optimized to make the transformed voltage V_t closer to the actual signal voltage V of the PVDF sensor. Therefore, the objective function and constraints for the optimization process are designed: the Sum of Squares for Errors (SSE) between the transformed voltage V_t through the equivalent electrical circuit model and the actual signal voltage V of the PVDF sensor is selected as the objective function:

$$SSE = \sum_{i=1}^n (V_i - V_{ti})^2 \quad (5)$$

The amplitude of the response derived from the equivalent electrical circuit model is subject to a non-negativity constraint, reflecting the

inherent positive of impulse force magnitudes in the underlying physical system. For example, the estimation results (in three different drop heights of 10 cm, 30 cm, and 50 cm) of the transformed force F_t from the signal voltage V show the applicability for different drop heights in Fig. 5a,b,c, respectively. The example with a drop height of 30 cm respectively demonstrates the model optimization results of the inversely solving method with the comparison between the transformed force F_t and actual impact force F , as shown in Fig. 6a and the SE of the optimization result is shown in Fig. 6b. As a consequence, the average error of the conversion is $MSE = 0.0179$, based on the optimization results of 50 sets of experimental data. Based on the above experimental results, the conversion from PVDF sensor output to impact force was ultimately achieved, which can be used for TBI assessment of focal brain damage.

3.2. Impact force analyze

During the experiments, the signal voltage V exhibits a waveform similar to a sine wave (Divided into a contact section and release section), which can be explained by the mechanical dynamics of the impact generation process, as shown in Fig. 7a. During the test, the dummy head is allowed to fall from different heights, so the PVDF sensor attached to the surface of the dummy head is compressed under the

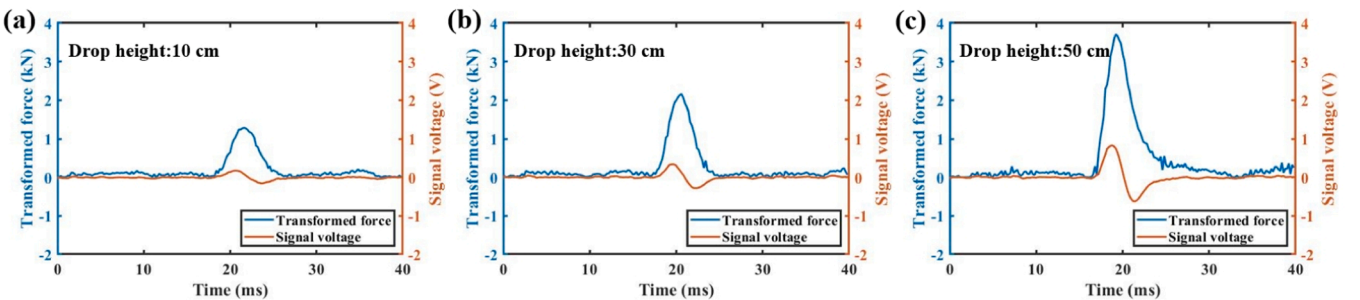


Fig. 5. Comparison of three examples between signal voltage V and transformed force F_t (a) drop height of 10 cm, (b) drop height of 30 cm, (c) drop height of 50 cm, respectively.

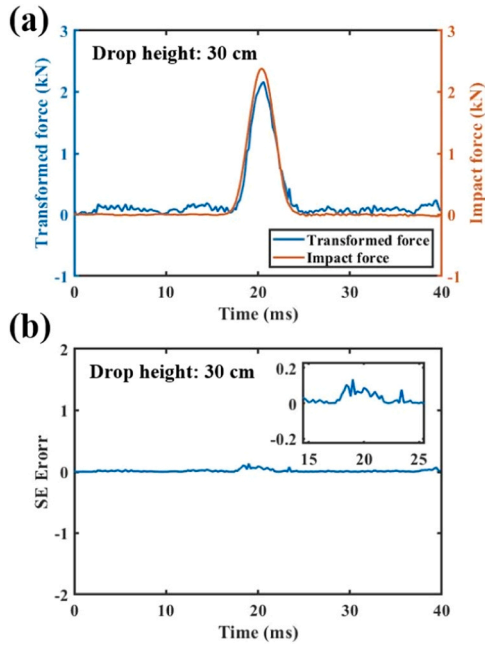


Fig. 6. (a) Comparison of an example (drop height of 30 cm) between impact force F and transformed force F_t . (b) SE error of an example (drop height of 30 cm) between impact force F and transformed force F_t .

impact, resulting in a positive voltage in the contact section. During the release section, the dummy head experiences a force opposite to gravity and rebounds, while PVDF film also experiences rebound after compression, resulting in a voltage opposite to the impact process. Besides, Fig. 7a also compares the signal voltage V of the dummy head dropping from different heights, and it can be quickly concluded that the peak value of signal voltage V is positively correlated with the drop height.

In the experiment, the PVDF film exhibits a transient, small-amplitude oscillatory response before stabilization attributed to the material's elasticity, which introduces residual signals in the transformed force F_t modeling, as evidenced by the data following the impulse signal in Fig. 7b. It is also apparent that when the drop height reaches 50 cm, the residual signal becomes pronounced due to the higher impact level, which leads to significant oscillation within the rebound of the head. Nevertheless, the influence of these residuals on the accuracy of model estimations is minimal during the impact contact process (as shaded in the enclosed portion of Fig. 7b). The waveform of the transformed force F_t (shown in Fig. 7b) behaves close to the impact force F , (shown in Fig. 7c), respectively, which is also the result of the model transformation. It is worth noting that compared to the impact

force F obtained by the quartz impact sensor, the signal fluctuation of the transformed force F_t after the impact process is more extensive. The underlying cause of this phenomenon is that the smart headband affixed to the dummy head generates relatively high-frequency vibrations during the impact process, whereas the quartz-based impact sensors fixed on the ground exhibit a more stable performance.

3.3. Estimation of impact acceleration

Although the impact force applied to the head can be directly measured, the acceleration of the head cannot be directly obtained by the PVDF film. Therefore, estimating the impact acceleration in TBI assessment is necessary based on the above evaluation result of impact force, which results in diffuse axonal injury or brain swelling evaluation applications. Firstly, The Pearson coefficient is used to evaluate the correlation between impact acceleration a and transformed force F_t (shown in Fig. 8a): the average Pearson correlation is 0.9463. Therefore, the correlation analysis results of the waveforms also provide support for the impact force obtained from the PVDF signal voltage to estimate impact acceleration. The impact acceleration a during the impact process is shown in Fig. 8b. Due to the accelerometer being fixed inside the dummy's head, the impact acceleration a (shown in Fig. 8b) contains more relevant characteristics of the impact process. In addition to the impact contact process displayed by the impact force, it also includes the acceleration oscillation signal after the dummy head rebounds, which is reflected in a more significant residual signal than the transformed force F_t (shown in Fig. 7b). Although the noise arising from the issues mentioned above is indeed challenging to evade during the measurement process, its effect on the impulsive phase (as shaded in the enclosed portion of Fig. 8b) is relatively minor and does not compromise the assessment of TBI through the sensor's measurement of impact force and the mapped acceleration.

To estimate the impact acceleration with impact force on the head through the output of the PVDF sensor, correlation analyses were performed between the peaks of the impact acceleration a and the impact force F , while between the impact acceleration a and transformed force F_t , respectively. To explore the correlation, a linear fitting method is used to solve the linear relationship between the above. Fig. 9 shows the peak scatter plots with the 95% confidence intervals for the correlation between peaks of the impact force F and peaks of the impact acceleration a (shown in Fig. 9a) and the correlation between peaks of the transformed force F_t and peaks of the impact acceleration a (shown in Fig. 9b), which were analyzed based on multiple impacts of the head dropping from different heights (10 cm, 20 cm, 30 cm, 40 cm, 50 cm). Next, based on the linear regression method, the relationships among the peaks of impact force F with impact acceleration a and transformed force F_t with impact acceleration a , respectively, are established based on the best-fitting line. R-squared (R^2 or the coefficient of determination) measures the goodness of fit between the data and the regression

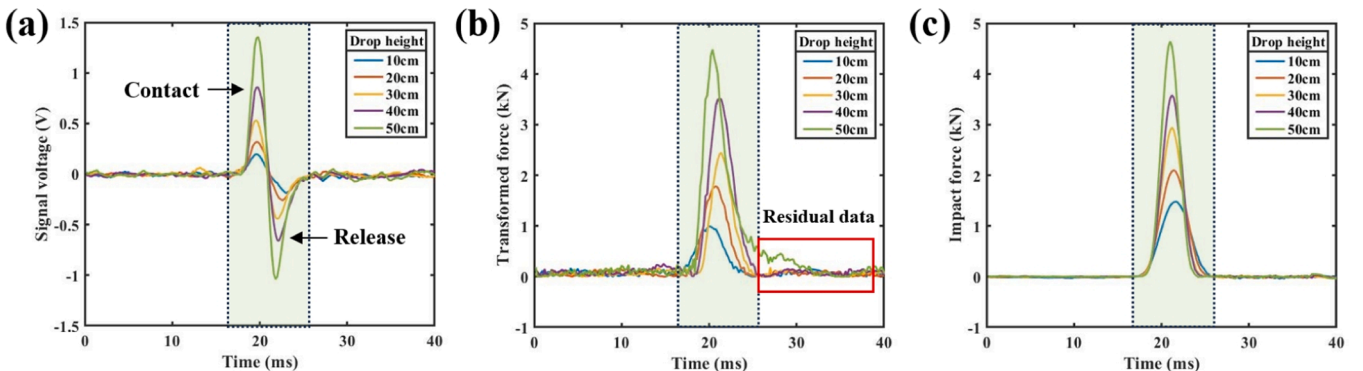


Fig. 7. Comparison of the time domain of different heights (10 cm, 20 cm, 30 cm, 40 cm, 50 cm) (a) signal voltage V ; (b) transformed force F_t ; (c) impact force F .

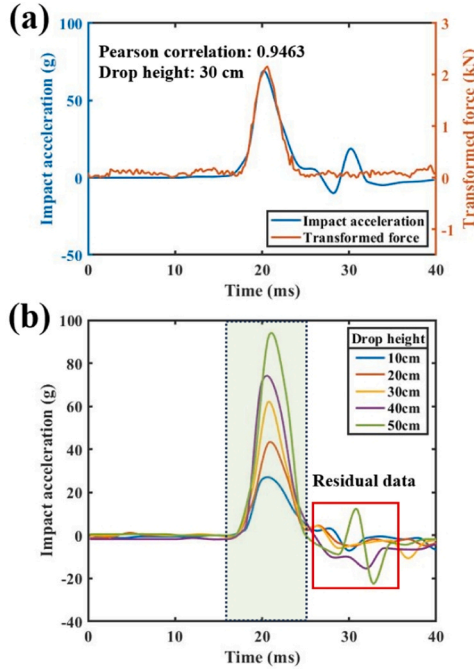


Fig. 8. (a) Comparison of an example (drop height of 30 cm) between impact acceleration a and transformed force F_t . (b) Comparison of impact acceleration a in the time domain of different heights (10 cm, 20 cm, 30 cm, 40 cm, 50 cm).

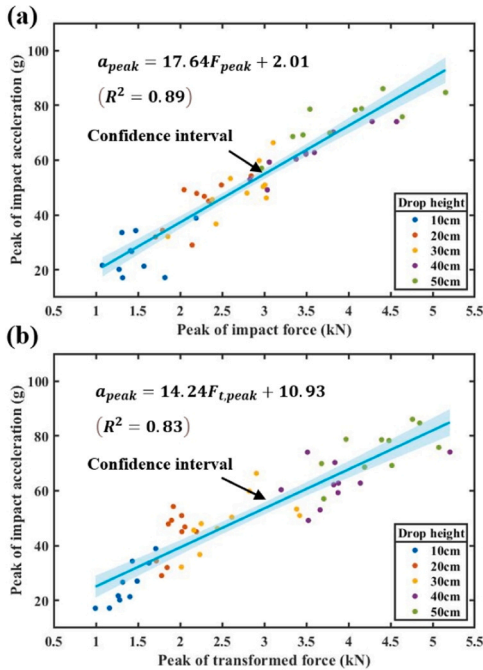


Fig. 9. (a) Peaks of the impact force F vs. Peaks of the impact acceleration a from different drop heights. (b) Peaks of the transformed force F_t vs. Peaks of the impact acceleration a from different drop heights.

model. The optimal fitting line determined by the least squares method and its R^2 are presented below:

$$\begin{aligned} a_{peak} &= 17.64F_{peak} + 2.01 \\ a_{peak} &= 14.24F_{t,peak} + 10.93 \end{aligned} \quad (6)$$

It can be seen from Fig. 9a that there is a positive correlation ($R^2=0.89$) between the peaks of the impact force F and the peaks of the

impact acceleration a , which is consistent with the conclusion in dynamic theory that force and acceleration are positively correlated. Besides, the peaks of transformed force F_t also show a positive correlation ($R^2=0.83$) with the peaks of impact acceleration a , as shown in Fig. 9b, which also verifies the result of the model transformation. In further analysis, while the correlation between transformed force F_t and impact acceleration a is relatively weaker compared to the association between impact force F and impact acceleration a , it still holds value in characterizing the nature of acceleration. Therefore, based on the aforementioned analytical results, the PVDF sensor proposed in this study not only derives impact forces through model transformation but also, as evidenced by correlation analyses, can characterize impact acceleration. In future work, we envision integrating impact force and acceleration within a smart headband to facilitate further real-time evaluation and assessment of TBI.

3.4. Wearable device enabling TBI assessment

In order to enable TBI assessment through the above impact evaluation method, relevant discussions were conducted on the wearable device system. According to practical application requirements, the PVDF-based smart headband must be integrated into wearable devices such as helmets to obtain real-time information on brain injuries. Data collection methods and damage identification play a role in the accuracy of the TBI assessment result in wearable device embedded systems. The data acquisition sampling rate should be based on the duration of the impulse process to ensure that the complete impulse time domain signal waveform can be obtained during the data acquisition process. According to the research of Li et al. [16] and the experimental results of this paper, it is proved that the time interval between impacts is generally more than 10 ms. Therefore, the acquisition speed of the embedded system can be set to the sampling rate of 5 kHz so that 50 data points can be obtained in the impact duration to meet the presentation of the complete waveform and for subsequent TBI analysis. To meet the sampling rate requirements, Arduino Uno was chosen as the microcontroller for this embedded system, with a clock speed of 16 MHz and a maximum ADC sampling rate of up to 10 kHz. Therefore, other microcontrollers that meet the sampling frequency requirements can be selected according to demand in practical applications. In addition, the microcontroller can also be connected to Bluetooth, storage, and localization modules for future medical applications of the wearable device. Due to the problem that a single sensor may not be able to detect some angular impact in the application of wearable devices in the real-life scenario, a multi-sensor array can be designed based on the method in this paper to meet the needs of multi-angle and multi-point collision. To meet the detection requirements of sensor arrays, data registers and multiplexers can be set to achieve separate acquisition of sensors at different locations, suggesting in [25]. Specifically, the multiplexer selects appropriate input channels based on the control signals of the microcontroller and routes the selected signal to the data registers. However, this method will reduce the system's sampling rate, and further research is needed on the data collection and storage methods.

Further, corresponding TBI evaluations can be conducted based on the impact force and acceleration obtained from the wearable devices mentioned above. According to the estimation method of impact force and impact acceleration information obtained from wearable devices proposed in this article, TBI for different brain injury situations can be evaluated separately. In order to accurately discriminate the severity of head injuries, standardized criterion such as Head Injury Criterion [26], a quantitative measure used in the field of injury biomechanics, can be used to assess the potential severity of head impacts and further estimate the TBI.

$$HIC = \max_{t_1, t_2} \left\{ \left[\frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \|\vec{a}(t)\| dt \right]^{2.5} (t_2 - t_1) \right\} \quad (7)$$

where $\|\vec{a}(t)\|$ is the magnitude of acceleration and the times bounded by $t_2 - t_1 < 15$ ms to maximize the value of HIC. Lower HIC values indicate a lower risk of severe head injury, whereas higher values suggest a greater risk. For instance, Hayes et al. [27] demonstrating that for a HIC score of 1000, there is a 16 % probability of a head injury while a HIC score of 1500 corresponds to a 56 % probability of a head injury. From Eq. 7, it can be seen that the impact acceleration, as a key parameter for calculating the HIC, plays an important role in the TBI assessment. In addition, for focal traumatic brain injury, biomechanical measures such as Cumulative Strain Damage Measure (CSDM) and Maximum Principal Strain (MPS) [28] can be used to assess the severity of focal TBI by quantifying the local tissue deformation and strain-induced damage within the brain caused by the external impact force on the head. In summary, the impact force evaluation and impact acceleration estimation method proposed in this article enables the embedded wearable device to accurately achieve impact detection, thereby further providing support for TBI evaluation based on impact force and acceleration.

4. Conclusion

This work is characterized by using the PVDF-based smart headband to evaluate and estimate the head impact force and acceleration for TBI assessment, respectively. In summary, we have designed and fabricated a smart headband based on a PVDF sensor, which can effectively evaluate the impact force and acceleration during the impact process on the head. To obtain signals of the impact process, experiments were conducted with a dummy head with a drop tower, which simulated the various impact levels of different intensities and accelerations on the head. Besides, a lumped parameter model for PVDF combining characteristics between the mechanical and electronic domain was established by a system identification method, which plays a significant role in evaluating the impact forces. According to the correlation analysis results, the acceleration can be estimated by the impact force on the head, which enables the TBI assessment of diffuse brain damage. The research above demonstrated that the smart headband can perceive different impact levels on the head, and thus, the wearable device has the potential to be used for head injury monitoring and assessment in various contact sports such as hockey, football, rugby, etc.

CRedit authorship contribution statement

Yunqi Cao: Writing – review & editing. **Dibo Hou:** Writing – review & editing. **Xuan Chen:** Software. **Haozhen Chi:** Data curation. **Haochuan Wan:** Writing – review & editing. **Wei Li:** Methodology. **Shuxun Wang:** Data curation. **Wangdi Du:** Writing – original draft, Conceptualization.

Declaration of Competing Interest

All authors disclosed no relevant relationships.

Data availability

The data that has been used is confidential.

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References

- [1] I. Moppett, Traumatic brain injury: assessment, resuscitation and early management, *Br. J. Anaesth.* 99 (1) (2007) 18–31.
- [2] D. Pavlovic, S. Pekic, M. Stojanovic, V. Popovic, Traumatic brain injury: neuropathological, neurocognitive and neurobehavioral sequelae, *Pituitary* 22 (2019) 270–282.
- [3] J.-Y. Jiang, G.-Y. Gao, J.-F. Feng, Q. Mao, L.-G. Chen, X.-F. Yang, J.-F. Liu, Y.-H. Wang, B.-H. Qiu, X.-J. Huang, Traumatic brain injury in China, *Lancet Neurol.* 18 (3) (2019) 286–295.
- [4] J.D. Cassidy, L. Carroll, P. Peloso, J. Borg, H. Von Holst, L. Holm, J. Kraus, V. Coronado, et al., Incidence, risk factors and prevention of mild traumatic brain injury: results of the who collaborating centre task force on mild traumatic brain injury, *J. Rehabil. Med.* 36 (0) (2004) 28–60.
- [5] A.I. Maas, D.K. Menon, P.D. Adelson, N. Andelic, M.J. Bell, A. Belli, P. Bragge, A. Brazinova, A. Buki, R.M. Chesnut, et al., Traumatic brain injury: integrated approaches to improve prevention, clinical care, and research, *Lancet Neurol.* 16 (12) (2017) 987–1048.
- [6] D.A. Patton, C.M. Huber, D. Jain, R.K. Myers, C.C. McDonald, S.S. Margulies, C. L. Master, K.B. Arbogast, Head impact sensor studies in sports: a systematic review of exposure confirmation methods, *Ann. Biomed. Eng.* 48 (2020) 2497–2507.
- [7] E. Le Flao, G.P. Siegmund, R. Borotkanics, Head impact research using inertial sensors in sport: a systematic review of methods, demographics, and factors contributing to exposure, *Sports Med.* (2022) 1–24.
- [8] C. Werner, K. Engelhard, Pathophysiology of traumatic brain injury, *Br. J. Anaesth.* 99 (1) (2007) 4–9.
- [9] K. Blennow, D.L. Brody, P.M. Kochanek, H. Levin, A. McKee, G.M. Ribbers, K. Yaffe, H. Zetterberg, Traumatic brain injuries, *Nat. Rev. Dis. Prim.* 2 (1) (2016) 1–19.
- [10] Y. Liu, A.G. Domel, S.A. Yousefsani, J. Kondic, G. Grant, M. Zeineh, D.B. Camarillo, Validation and comparison of instrumented mouthguards for measuring head kinematics and assessing brain deformation in football impacts, *Ann. Biomed. Eng.* 48 (2020) 2580–2598.
- [11] N.J. Cecchi, D.C. Monroe, T.J. Oros, S.L. Small, J.W. Hicks, Laboratory evaluation of a wearable head impact sensor for use in water polo and land sports, *Proc. Inst. Mech. Eng. Part P: J. Sports Eng. Technol.* 234 (2) (2020) 162–169.
- [12] K.L. O'Connor, S. Rowson, S.M. Duma, S.P. Broglio, Head-impact-measurement devices: a systematic review, *J. Athl. Train.* 52 (3) (2017) 206–227.
- [13] Y. Huang, X. Fan, S.-C. Chen, N. Zhao, Emerging technologies of flexible pressure sensors: materials, modeling, devices, and manufacturing, *Adv. Funct. Mater.* 29 (12) (2019) 1808509.
- [14] S.J. Rupitsch, Piezoelectric sensors and actuators, *Simulation of Piezoelectric Sensor and Actuator Devices*, In: *Piezoelectric Sensors and Actuators*. Topics in Mining, Metallurgy and Materials Engineering (2019) 83–126.
- [15] A. Khan, Z. Abas, H.S. Kim, I.-K. Oh, Piezoelectric thin films: an integrated review of transducers and energy harvesting, *Smart Mater. Struct.* 25 (5) (2016) 053002.
- [16] Q. Li, X. Liao, X. Huang, X. Wei, X. Zhang, Impact resistance test system for the helmet based on a polyvinylidene fluoride piezoelectric sensor array, *Int. J. Occup. Saf. Ergon.* 29 (1) (2023) 199–206.
- [17] W. Li, D. Torres, T. Wang, C. Wang, N. Sepulveda, Flexible and biocompatible polypropylene ferroelectric nanogenerator (feng): on the path toward wearable devices powered by human motion, *Nano Energy* 30 (2016) 649–657.
- [18] J. Lekkala, J. Tuppurainen, M. Paajanen, Material and operational properties of large-area membrane type sensors for smart environments, In: *Proceedings of the XVII IMEKO World Congress*, 2003, pp. 2045–2048.
- [19] R.H. Brown, Piezo film: form and function, *Sens. Actuators A: Phys.* 22 (1–3) (1990) 729–733.
- [20] F. Hernandez, L.C. Wu, M.C. Yip, K. Laksari, A.R. Hoffman, J.R. Lopez, G.A. Grant, S. Kleiven, D.B. Camarillo, Six degree-of-freedom measurements of human mild traumatic brain injury, *Ann. Biomed. Eng.* 43 (2015) 1918–1934.
- [21] M. Smith, S. Kar-Narayan, Piezoelectric polymers: theory, challenges and opportunities, *Int. Mater. Rev.* 67 (1) (2022) 65–88.
- [22] S. Kaiki, J. Lekkala, A lumped-parameter transducer model for piezoelectric and ferroelectric polymers, *Measurement* 45 (3) (2012) 453–458.
- [23] H.-P. Lv, W.-Q. Liao, Y.-M. You, R.-G. Xiong, Inch-size molecular ferroelectric crystal with a large electromechanical coupling factor on par with barium titanate, *J. Am. Chem. Soc.* 144 (48) (2022) 22325–22331.
- [24] L. Liu, Q. Li, H. Yun, J. Liang, X. Ma, Composite modeling and parameter identification of broad bandwidth hysteretic dynamics in piezo-electric fast steering platform, *Mech. Syst. Signal Process.* 121 (2019) 97–111.
- [25] Zafeirakis I., Filippidou M.K., Chatzandroulis S., Kyriakis-Bitzaros E.D., Stathopoulos N., Vassiliadis S. Design and implementation of a re-configurable embedded system for capacitive sensor array interface, In: *Proceedings of the 2018 7th International Conference on Modern Circuits and Systems Technologies (MOCAS)* (2018) 1–4. IEEE.
- [26] F. Hernandez, L.C. Wu, M.C. Yip, K. Laksari, A.R. Hoffman, J.R. Lopez, G.A. Grant, S. Kleiven, D.B. Camarillo, Six degree-of-freedom measurements of human mild traumatic brain injury, *Ann. Biomed. Eng.* 43 (2015) 1918–1934.
- [27] W.C. Hayes, M.S. Erickson, E.D. Power, Forensic injury biomechanics, *Annu. Rev. Biomed. Eng.* 15 (9) (2007) 55–86.
- [28] Takhouas E.G., Craig M.J., Moorhouse K., McFadden J., Hasija V. Development of brain injury criteria (BrIC). SAE Technical Paper, 11 (2013).

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